Important Considerations in the Use of Regression Equation:

1. Regression equation exhibits only the relationship between the respective two variables. Cause and effect study shall not be carried out using regression analysis.
2. The regression equation is fitted to the given values of the independent variable. Hence, the fitted equation can be used for prediction purpose corresponding to the values of the regressor within its range. Interpolation of values of the response variable may be done corresponding to the values of the regressor from its range only. The results obtained from extrapolation work could not be interpreted.

Example 5.1

Construct the simple linear regression equation of Y on X if $n = 7, \sum_{i=1}^{n} x_i = 113$, $\sum_{i=1}^{n} x_i^2 = 1983$, $\sum_{i=1}^{n} y_i = 182$ and $\sum_{i=1}^{n} x_i y_i = 3186$.

Solution:

The simple linear regression equation of Y on X to be fitted for given data is of the form

$$\hat{Y} = a + bx \tag{1}$$

The values of 'a' and 'b' have to be estimated from the sample data solving the following normal equations.

$$na + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \tag{2}$$

$$a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i \tag{3}$$

Substituting the given sample information in (2) and (3), the above equations can be expressed as

$$7a + 113b = 182 \tag{4}$$

$$113a + 1983b = 3186 \tag{5}$$

$$(4) \times 113 \Rightarrow 791a + 12769b = 20566$$

$$(5) \quad \times 7 \Rightarrow 791a + 13881b = 22302$$

$$(-) \quad (-) \quad (-)$$

$$-1112b = -1736$$

$$\Rightarrow b = \frac{1736}{1112} = 1.56$$

$$b = 1.56$$

Substituting this in (4) it follows that,

$$7a + 113 \times 1.56 = 182$$

$$7a + 176.28 = 182$$

$$7a = 182 - 176.28$$

$$= 5.72$$

Hence, $a = 0.82$

12th_Statistics_EM_Unit_5.indd 135 1/28/2020 6:36:31 PM